



U.S. Department  
of Transportation

**Pipeline and Hazardous  
Materials Safety  
Administration**

**July 21, 2023**

1200 New Jersey Avenue, SE  
Washington, DC 20590

DOT-SP 21229  
FIRST REVISION

**EXPIRATION DATE: 2027-06-30**

(FOR RENEWAL, SEE 49 CFR § 107.109)

1. GRANTEE: Mercedes-Benz U.S. International, Inc.  
Vance, AL
2. PURPOSE AND LIMITATION:
  - a. This special permit authorizes the transportation in commerce of certain lithium ion battery assemblies that exceed 35 kg aboard cargo-only aircraft. This special permit provides no relief from the Hazardous Materials Regulations (HMR) or the International Civil Aviation Organization's Technical Instructions for the Safe Transport of Dangerous Goods by Air (ICAO TI) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
  - b. The safety analyses performed in the development of this special permit only considered the hazards and risks associated with the transportation in commerce.
  - c. No party status will be granted to this special permit.
  - d. This special permit serves as an approval under ICAO TI Special Provision A99 and as a "Competent Authority Approval" as defined under 49 CFR § 107.1.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180 and the ICAO TI.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 172.101 Hazardous Materials Table Column (9B) in that lithium batteries may not exceed 35 kg and 49 CFR 173.185(b)(6) in that transportation of large UN packagings aboard aircraft is not permitted, except as specified herein.

Tracking Number: 2023054085

5. BASIS: This special permit is based on the application of Mercedes-Benz U.S. International, Inc. dated May 1, 2023, and submitted in accordance with § 107.109.
6. HAZARDOUS MATERIALS (49 CFR 172.101):

| Hazardous Materials Description                                      |                        |                       |               |
|----------------------------------------------------------------------|------------------------|-----------------------|---------------|
| Proper Shipping Name                                                 | Hazard Class/ Division | Identification Number | Packing Group |
| Lithium ion batteries <i>including lithium ion polymer batteries</i> | 9                      | UN3480                | N/A           |

\*Only the lithium ion battery assemblies specified in the April 23, 2021 application on file with the Office of Hazardous Materials Safety (OHMS) or the battery assemblies of the same tested type may be transported under the terms of this special permit.

7. SAFETY CONTROL MEASURES:

a. OPERATIONAL CONTROLS:

- (1) Cells and battery assemblies must be protected against short-circuiting (such as capping of battery assembly terminals).
- (2) The battery assemblies must be equipped with an effective means to prevent dangerous reverse current flow when cells or battery assemblies are connected in parallel.
- (3) The battery assemblies must be offered for transportation at a state of charge not greater than 30%.
- (4) The battery assembly must be equipped with a Battery Management System capable of monitoring the battery assembly and preventing short circuits, or over discharge between the modules in the assembly and any overheat or overcharge of the battery assembly.

b. TESTING:

- (1) All lithium ion cells and battery assemblies must be of a type that have passed all required tests as specified in the "UN Manual of Tests and Criteria, 6<sup>th</sup> Revised Edition."

(2) The battery assemblies not equipped with overcharge protection and not designed for use as a component in equipment that affords such protection must have passed the overcharge test (T.7) as specified in the “UN Manual of Tests and Criteria, 6<sup>th</sup> Revised Edition.”

c. PACKAGING:

(1) Inner Packaging: The battery assembly must be placed into a poly bag and then bolted down through the bag, which must be surrounded with non-combustible, non-conductive cushioning material on all sides.

(2) Outer Packaging: The inner packaging must be placed in a UN Standard 50D Large Packaging tested to the Packing Group I or II performance standard.

(3) The packaging must comply with Packing Instruction 974 of the Supplement to the ICAO TI. Additionally, waste lithium battery assemblies and lithium battery assemblies being shipped for recycling or disposal are forbidden from air transport, and lithium ion battery assemblies must not be packed in the same outer packaging with substances and articles of Class 1 (explosives) other than Division 1.4S, Division 2.1 (flammable gases), Class 3 (flammable liquids), Division 4.1 (flammable solids), or Division 5.1 (oxidizers).

(4) Only one battery assembly may be packaged in an outer packaging. The net weight of the battery assembly may not exceed that specified in the April 23, 2021, application on file with OHMS.

d. MARKING REQUIREMENTS: Each package, and overpack if used, prepared under the provisions of this special permit must be plainly marked with the special permit number: DOT-SP 21229.

8. SPECIAL PROVISIONS:

a. Under the terms of this special permit, the grantee may only offer hazardous materials (i.e., the grantee is not authorized as a carrier).

b. A person who is not a holder of this special permit who receives a package covered by this special permit may reoffer it for transportation provided no modification or change is made to the package and it is reoffered for transportation in conformance with this special permit, the HMR, and the ICAO TI.

c. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

d. This special permit in no way affects the need to obtain any required authorizations from other agencies of the United States Government or from the competent authorities of countries of origin, transit, and destination and the State of the Operator.

e. The special permit holder must maintain a record of all activity conducted under the authority granted in this special permit. The record must contain a complete listing and number of shipments made to include and upon request make this information available to a DOT representative or an enforcement official. The record must contain a listing and number of shipments made to include:

- (1) Dates of shipment; and
- (2) Description of each type of shipment.

9. MODES OF TRANSPORTATION AUTHORIZED: Cargo-only aircraft.

10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each aircraft used to transport packages covered by this special permit. The shipper must furnish a copy of this special permit to the air carrier before or at the time the shipment is tendered.

11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:

- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)—“The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005),

amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. **REPORTING REQUIREMENTS:** Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover  
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Material Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: Jephthah Nti